Parallel Algorithms for Parallel Computing

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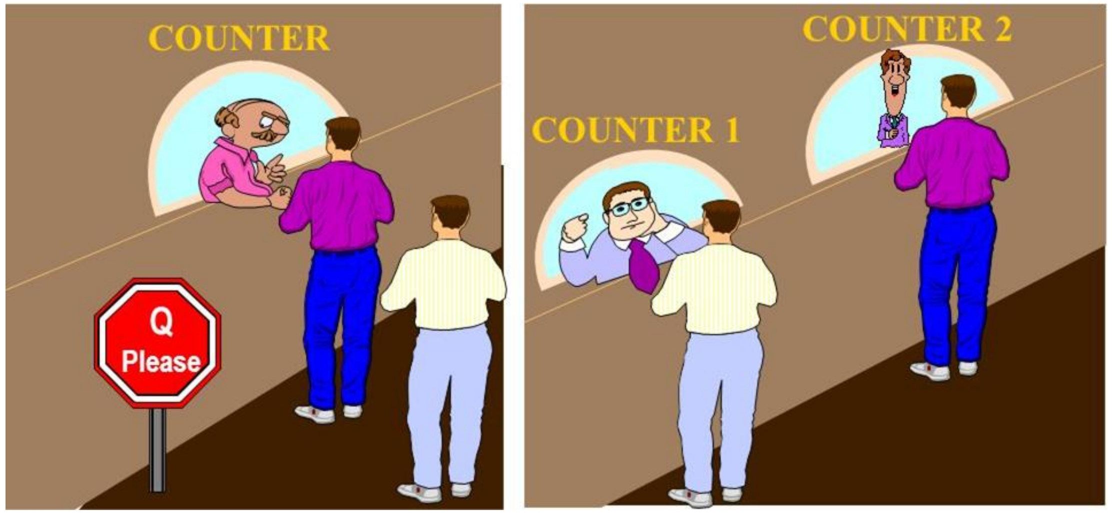


OUTLINE

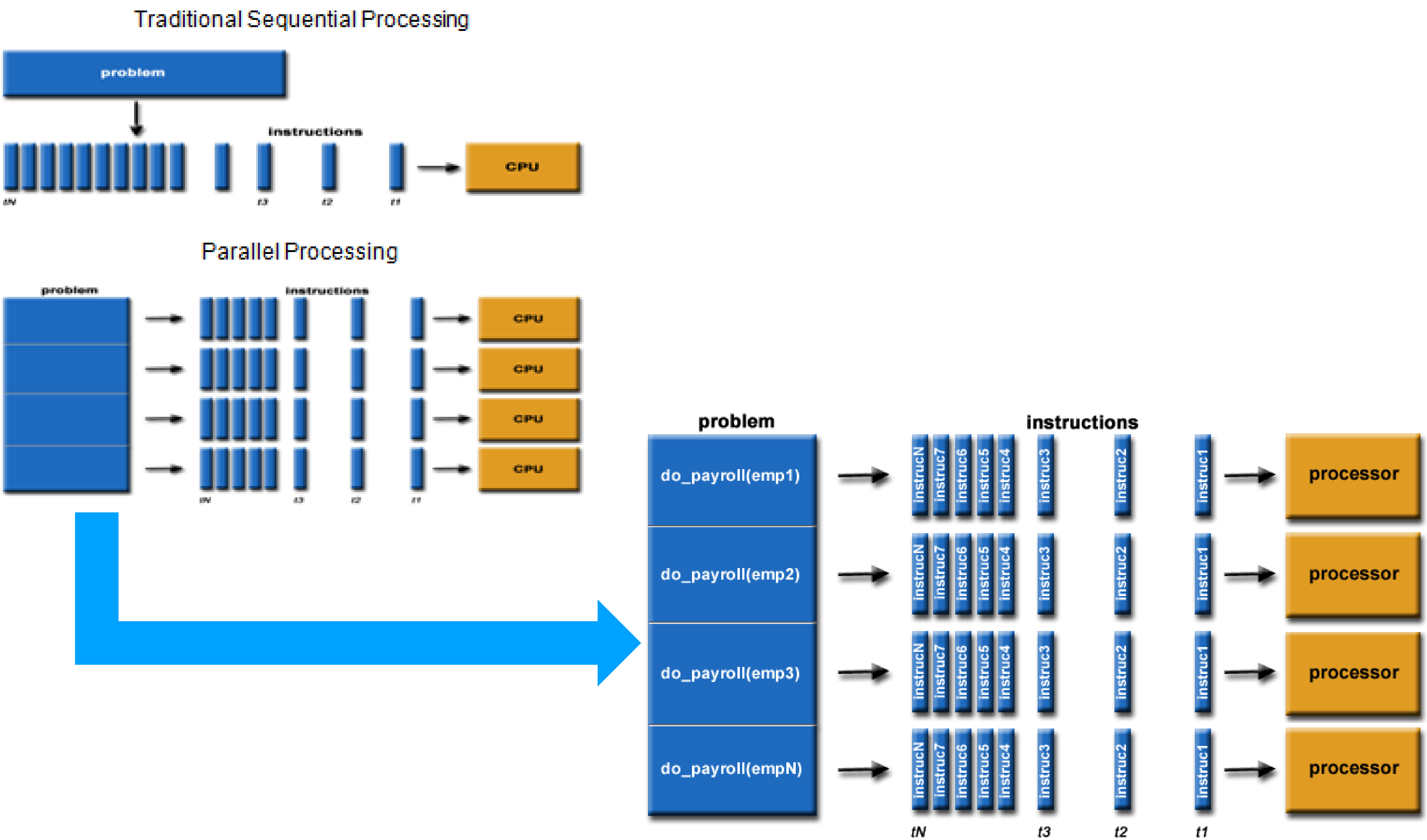
- Why Parallel and Distributed Computing Curriculum?
- What is Parallel Computing – videos
- Unplugged activity : More Processors are Not Always the Best
    - Activity phases: sequential and parallel
    - Group/class discussion
    - Survey

Why a Parallel and Distributed Computing Curriculum?

An initiative mandated by the National Science Foundation (NSF) and the ACM/IEEE Computer Society

https://grid.cs.gsu.edu/~tcpp/curriculum/?q=curr_whypdc.html

What is Parallel Computing?



What is Parallel Computing?

- Digging Holes: https://youtu.be/bun_WSB9iRw?si=TOVqmIKx6OEmXmnd
- On a more serious note: https://www.youtube.com/watch?v=q7sgzDH1cR8
- More video for later: https://www.youtube.com/watch?v=GXrl3oJ3Eak
- Detailed explanations: https://computing.llnl.gov/tutorials/parallel_comp/#Abstract

Unplugged activity

More processors are not always the best

Unplugged activity 1 description

The activity involves participation of two or more students for the following phases:

Uniprocessor Sequential phase

Participant 1 - acts as a processor to perform

Task 1: writing a sentence (**more processors are not always the best**), followed by

Task 2: assigning a numeric value as an index (**0,1,...32**) to each character in the written sentence.

Participant 2 - times the tasks performed by participant 1.

Unplugged activity 1 description cont'd...

An example of the two writing tasks:

M	o	r	e	P	r	o	c	e	s	s	o	r	s	A	r	e	N	o	t	A	l	w	a	y	s	T	h	e	B	e	s	t
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32

The above two tasks can be timed either in serial or in parallel in one of the activity phases.

NOTE: all processors (participants) are provided their own local buffer to work on. Thus, all processors (participants will work on their own sheets of paper).

Unplugged activity 1 description cont'd...

Uniprocessor Multi Tasking Phase

Participant 1 - acts as a processor to perform

Task 1: writing a sentence (**more processors are not always the best**), interleaved character by character with

Task 2: assigning a numeric value as an index (**0,1,...32**) to each character in the written sentence, interleaved with Task 1, number by number

Participant 2 - times the tasks performed by participant 1.

Compares performance with Sequential Phase

Unplugged activity 1 description cont'd...

Two Processor Parallel Phase

Participant 1 - acts as a parallel processor to perform

Task 1: writing a sentence (**more processors are not always the best**)

Participant 2 - acts as a parallel processor to perform

Task 2: assigning a numeric value as an index (**0,1,...32**) to each character in the written sentence. Starting and ending index values are given. Starts at the same time as Participant 1.

Participant 3 - times the tasks performed by parallel participants 1 and 2. Overall time is calculated as the time of the slower processor.

Compares performance with Sequential Phase and multi-tasking phase

Unplugged activity 1 description cont'd...

Multi Processor Parallel Phase

Participant 1 -

|(Task 1)/2|: writing a partial sentence (**more processors are**)

Participant 2 -

|(Task 1)/2|: writing a partial sentence (**not always the best**), starts at the same time as participant 1.

Participant 3 -

|(Task 2)/2|: assigning a numeric value as an index (**starting at 0**) to each character in the partial sentence written by participant 1. Starts with participant 1. Once done sends last index + 1 value to participant 4.

Participant 4 -

|(Task 2)/2|: assigning a numeric value as an index (**continues from participant 3**) to each character in the partial sentence written by participant 2. Gets starting index from participant 3. **Overhead:** may have to wait on participant 3 to continue proper indexing.

Participant 5 - times the tasks performed by parallel participants. Overall time is calculated as the time of the slowest processor

Compares performance with Sequential Phase and multi-tasking phase and previous parallel scenario.

Amdahl's Law limitations: Scenario on Previous slide
A suggested Improvement - Increase # of TASKS

Participant 1 –

|(Task 1)/2|: writing a partial sentence (more processors are not always the best)

Participant 2 –

|(Task 1)/2|: writing a partial sentence (more processors are not always the best), starts at the same time as participant 1.

Participant 3 –

Task 2: assigning a numeric value as an index to each character in the sentence written by participants 1 and 2. Starting and ending index are provided.

Participant 4 -

Task 3: calculating the total number of characters for each word in the sentence

Participant 5 – times the tasks performed by parallel participants. Overall time is calculated as the time of the slowest processor

Compares performance with Sequential Phase and multi-tasking phase and previous parallel scenario

Group Discussion:
what did we learn?

Survey link

Thank you for your participation!

Hope you enjoyed the parallelism day